Home Credit Default Risk — Exploratory Data Analysis

```
library(data.table)
library(dplyr)
library(ggplot2)
library(scales)

train <- fread("../data/raw/application_train.csv")
test  <- fread("../data/raw/application_test.csv")
```

Business Problem

Home Credit provides unsecured consumer loans to applicants who traditional banks often turn away, including many with little or no formal credit history. This project builds a model to predict, at the time of application, how likely an applicant is to repay, in order to reduce default losses and expand responsible lending to thin-file applicants. This notebook explores `application_train.csv` and `application_test.csv` to understand the data before any modeling work begins.

Standing Checks

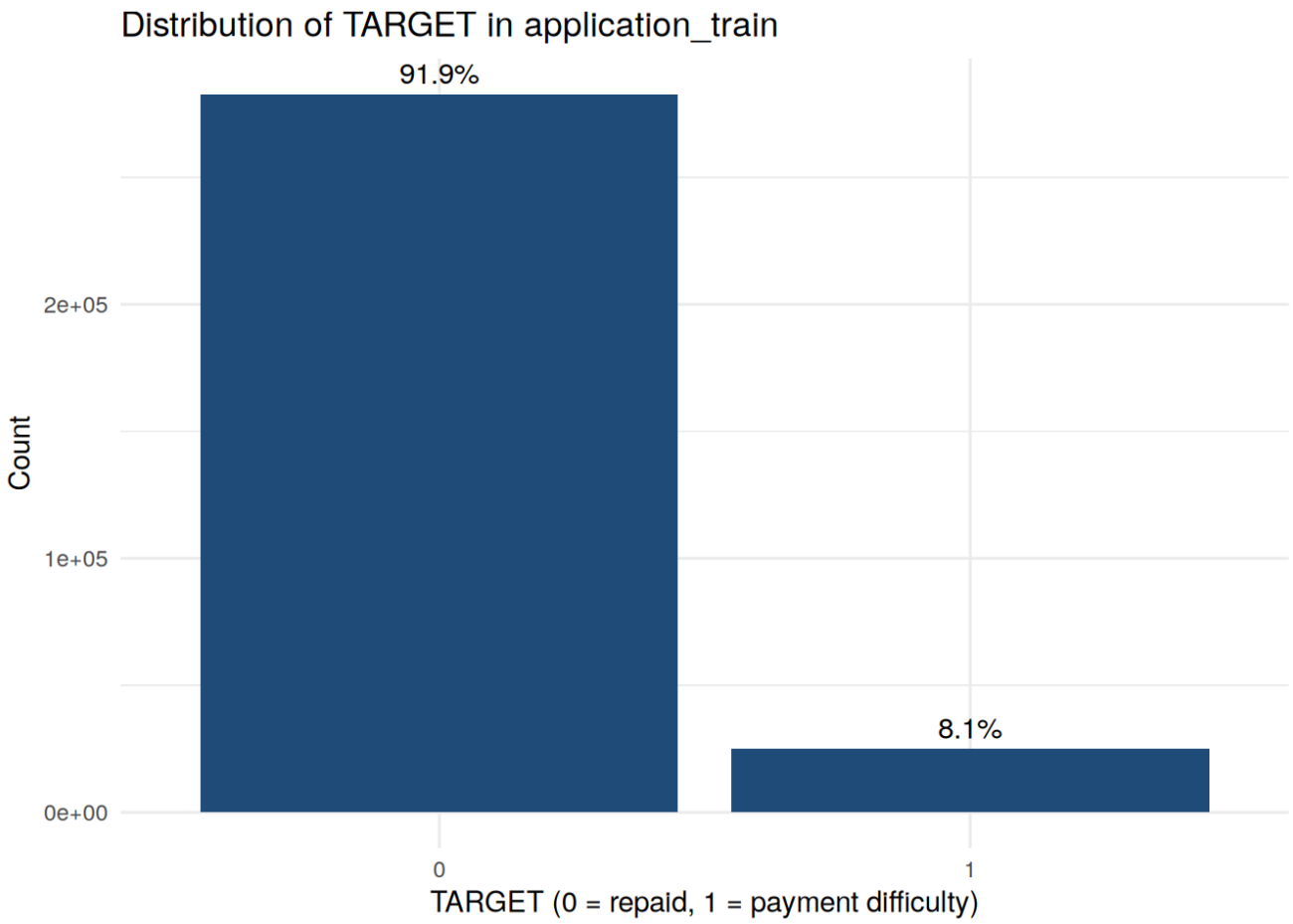
S1: Target Distribution

```
target_dist <- train %>%
  count(TARGET) %>%
  mutate(pct = n / sum(n) * 100)

target_dist
```

```
   TARGET      n       pct
1:      0 282686 91.927118
2:      1  24825  8.072882
```

```
ggplot(target_dist, aes(x = factor(TARGET), y = n)) +
  geom_col(fill = "#1F4E79") +
  geom_text(aes(label = paste0(round(pct, 1), "%")), vjust = -0.5) +
  labs(title = "Distribution of TARGET in application_train",
       x = "TARGET (0 = repaid, 1 = payment difficulty)",
       y = "Count") +
  theme_minimal()
```



TARGET=0 accounts for 282,686 rows (91.9%) and TARGET=1 accounts for 24,825 rows (8.1%) of `application_train`.

S2: Missingness and Relationship to TARGET

```
missing_pct <- sapply(train, function(x) mean(is.na(x)) * 100)
missing_tbl <- data.frame(column = names(missing_pct), pct_missing = missing_pct) %>
  filter(pct_missing > 0) %>%
  arrange(desc(pct_missing))

head(missing_tbl, 15)
```

	column	pct_missing
COMMONAREA_AVG	COMMONAREA_AVG	69.87230
COMMONAREA_MODE	COMMONAREA_MODE	69.87230
COMMONAREA_MEDI	COMMONAREA_MEDI	69.87230
NONLIVINGAPARTMENTS_AVG	NONLIVINGAPARTMENTS_AVG	69.43296
NONLIVINGAPARTMENTS_MODE	NONLIVINGAPARTMENTS_MODE	69.43296
NONLIVINGAPARTMENTS_MEDI	NONLIVINGAPARTMENTS_MEDI	69.43296
LIVINGAPARTMENTS_AVG	LIVINGAPARTMENTS_AVG	68.35495
LIVINGAPARTMENTS_MODE	LIVINGAPARTMENTS_MODE	68.35495

```
LIVINGAPARTMENTS_MEDI       LIVINGAPARTMENTS_MEDI    68.35495
FLOORSMIN_AVG                       FLOORSMIN_AVG    67.84863
FLOORSMIN_MODE                     FLOORSMIN_MODE    67.84863
FLOORSMIN_MEDI                     FLOORSMIN_MEDI    67.84863
YEARS_BUILD_AVG                   YEARS_BUILD_AVG    66.49778
YEARS_BUILD_MODE                 YEARS_BUILD_MODE    66.49778
YEARS_BUILD_MEDI                 YEARS_BUILD_MEDI    66.49778
```

```r
nrow(missing_tbl)
```

```
[1] 61
```

```r
train_flag <- train %>%
  mutate(ext_source_missing_any = is.na(EXT_SOURCE_1) | is.na(EXT_SOURCE_2) | is.na(

train_flag %>%
  group_by(ext_source_missing_any) %>%
  summarise(n = n(), target_rate = mean(TARGET) * 100)
```

```
# A tibble: 2 x 3
  ext_source_missing_any      n target_rate
  <lgl>                   <int>       <dbl>
1 FALSE                  109589        7.30
2 TRUE                   197922        8.50
```

61 columns in `application_train` have at least some missing values, with the highest-missingness columns being housing/building-detail fields near 70% missing; applicants missing at least one EXT_SOURCE score have a TARGET rate of 8.50% compared to 7.30% for those with all three present.

S3: Impossible or Sentinel-Like Values (Table-Wide)

```r
numeric_cols <- train %>% select(where(is.numeric)) %>% select(-SK_ID_CURR, -TARGET)

neg_check <- sapply(numeric_cols, function(x) any(x < 0, na.rm = TRUE))
neg_cols <- names(neg_check[neg_check])

days_employed_max <- max(train$DAYS_EMPLOYED, na.rm = TRUE)
days_employed_max_count <- sum(train$DAYS_EMPLOYED == days_employed_max, na.rm = TRU

gender_vals <- train %>% count(CODE_GENDER)
gender_vals
```

```
   CODE_GENDER      n
1:           F 202448
```

```
2:           M 105059
3:         XNA      4
```

```
neg_cols
```

```
[1] "DAYS_BIRTH"             "DAYS_EMPLOYED"          "DAYS_REGISTRATION"
[4] "DAYS_ID_PUBLISH"        "DAYS_LAST_PHONE_CHANGE"
```

```
days_employed_max
```

```
[1] 365243
```

```
days_employed_max_count
```

```
[1] 55374
```

5 numeric columns contain negative values (`DAYS_BIRTH`, `DAYS_EMPLOYED`, `DAYS_REGISTRATION`, `DAYS_ID_PUBLISH`, `DAYS_LAST_PHONE_CHANGE` — all day-offsets from the application date, negative by convention); `DAYS_EMPLOYED` has a maximum value of 365243, occurring in 55,374 rows, which is a positive value inconsistent with the negative day-offset convention used elsewhere in the same column; `CODE_GENDER` includes a category labeled “XNA” (4 rows) in addition to “M” and “F”.

S4: Grain, Primary Key, and Duplicates

```
nrow(train)
```

```
[1] 307511
```

```
length(unique(train$SK_ID_CURR)) == nrow(train)
```

```
[1] TRUE
```

```
nrow(test)
```

```
[1] 48744
```

```
length(unique(test$SK_ID_CURR)) == nrow(test)
```

```
[1] TRUE
```

```
length(intersect(train$SK_ID_CURR, test$SK_ID_CURR))
```

```
[1] 0
```

`application_train` has 307,511 rows with `SK_ID_CURR` unique across all of them; `application_test` has 48,744 rows with `SK_ID_CURR` also unique across all of them; there are 0 `SK_ID_CURR` values shared between the two tables.

S5: Temporal Direction

```
day_cols <- names(train)[grepl("^DAYS_", names(train))]
day_cols
```

```
[1] "DAYS_BIRTH"             "DAYS_EMPLOYED"          "DAYS_REGISTRATION"
[4] "DAYS_ID_PUBLISH"        "DAYS_LAST_PHONE_CHANGE"
```

```
train %>% select(all_of(day_cols)) %>% summary()
```

```
  DAYS_BIRTH     DAYS_EMPLOYED    DAYS_REGISTRATION DAYS_ID_PUBLISH
 Min.   :-25229   Min.   :-17912   Min.   :-24672    Min.   :-7197
 1st Qu.:-19682   1st Qu.: -2760   1st Qu.: -7480    1st Qu.:-4299
 Median :-15750   Median : -1213   Median : -4504    Median :-3254
 Mean   :-16037   Mean   : 63815   Mean   : -4986    Mean   :-2994
 3rd Qu.:-12413   3rd Qu.:  -289   3rd Qu.: -2010    3rd Qu.:-1720
 Max.   : -7489   Max.   :365243   Max.   :     0    Max.   :    0

 DAYS_LAST_PHONE_CHANGE
 Min.   :-4292.0
 1st Qu.:-1570.0
 Median : -757.0
 Mean   : -962.9
 3rd Qu.: -274.0
 Max.   :    0.0
 NA's   :1
```

Five columns begin with `DAYS_` (`DAYS_BIRTH`, `DAYS_EMPLOYED`, `DAYS_REGISTRATION`, `DAYS_ID_PUBLISH`, `DAYS_LAST_PHONE_CHANGE`), and all are framed as offsets relative to the application date rather than absolute dates, which is consistent with being decision-time information; without the data dictionary, it cannot be confirmed whether any column reflects information recorded after the application decision.

S6: Train/Test Consistency in Columns and Category Levels

```
setdiff(names(train), names(test))
```

```
[1] "TARGET"
```

```r
setdiff(names(test), names(train))
```

```
character(0)
```

```r
char_cols <- train %>% select(where(is.character)) %>% names()

level_diffs <- lapply(char_cols, function(col) {
  train_levels <- unique(train[[col]])
  test_levels <- unique(test[[col]])
  only_in_train <- setdiff(train_levels, test_levels)
  only_in_test <- setdiff(test_levels, train_levels)
  data.frame(column = col,
             only_in_train = paste(only_in_train, collapse = "; "),
             only_in_test = paste(only_in_test, collapse = "; "))
})

level_diffs_df <- do.call(rbind, level_diffs)
level_diffs_df %>% filter(only_in_train != "" | only_in_test != "")
```

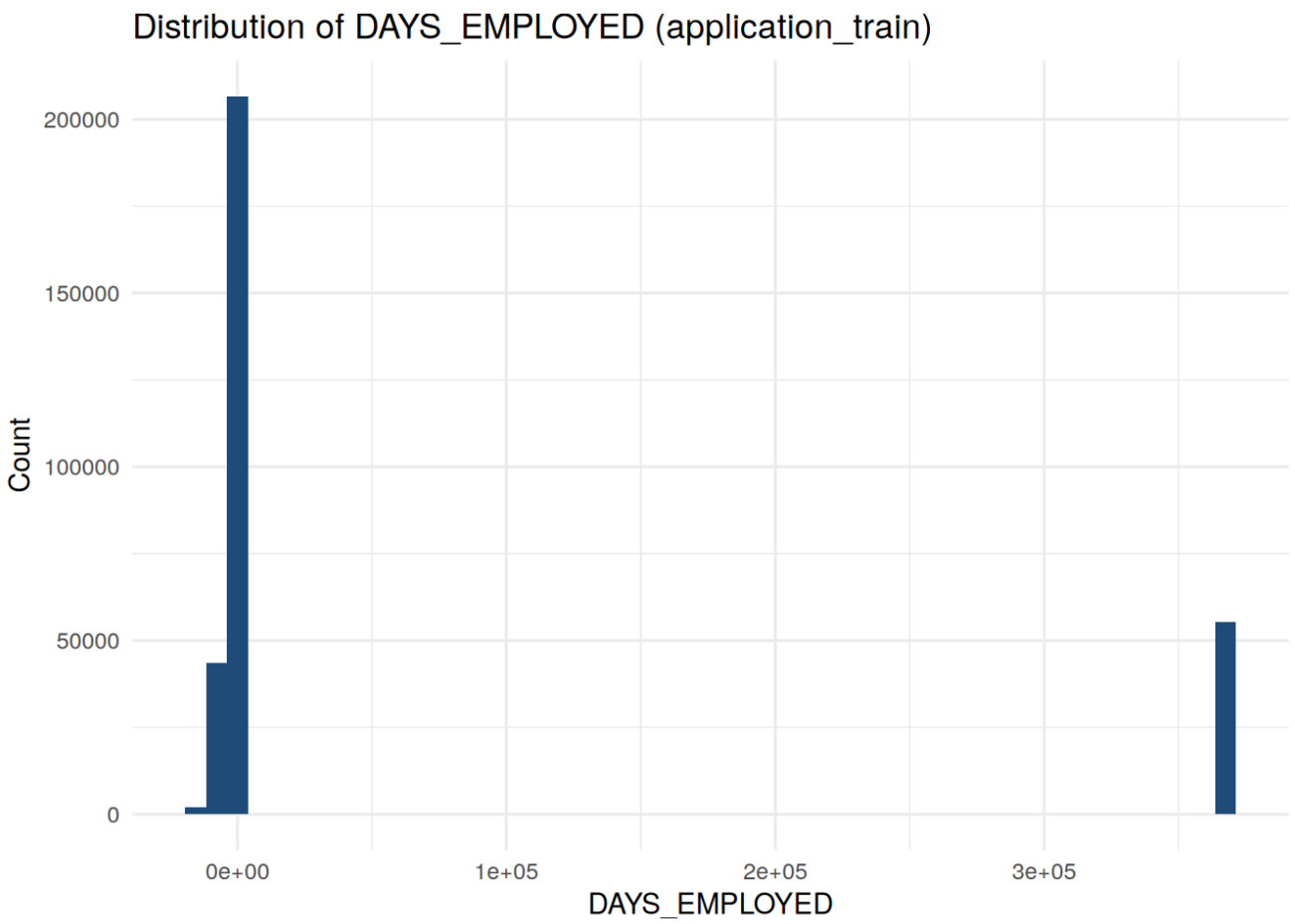
```
              column   only_in_train only_in_test
1        CODE_GENDER             XNA
2   NAME_INCOME_TYPE Maternity leave
3 NAME_FAMILY_STATUS         Unknown
```

The only column present in `application_train` and absent from `application_test` is `TARGET`; among categorical columns, `CODE_GENDER`'s "XNA" category, `NAME_INCOME_TYPE`'s "Maternity leave" category, and `NAME_FAMILY_STATUS`'s "Unknown" category appear in `application_train` but not in `application_test`.

EDA Questions

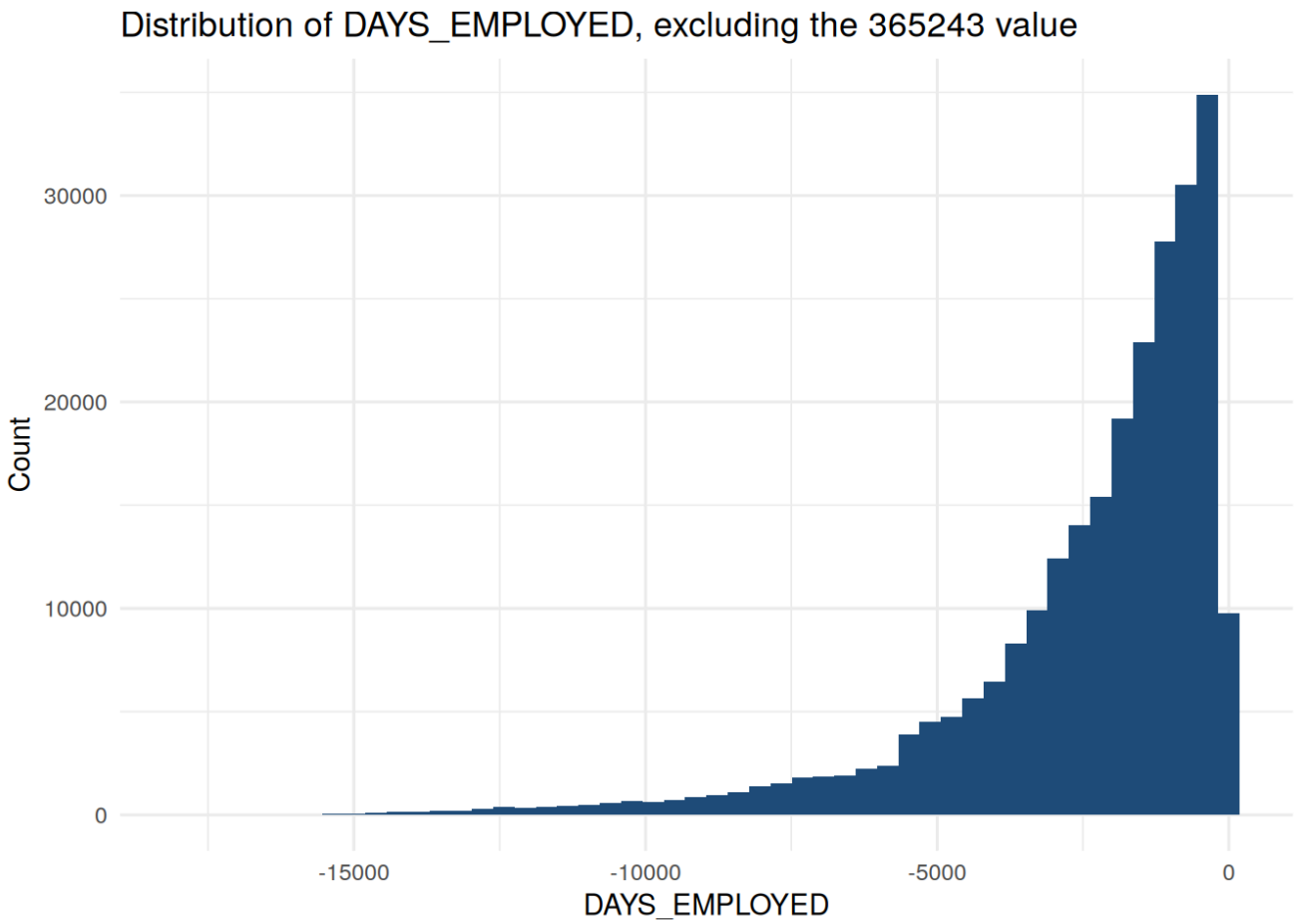
Question 8: Distribution of DAYS_EMPLOYED

```r
ggplot(train, aes(x = DAYS_EMPLOYED)) +
  geom_histogram(bins = 50, fill = "#1F4E79") +
  labs(title = "Distribution of DAYS_EMPLOYED (application_train)",
       x = "DAYS_EMPLOYED", y = "Count") +
  theme_minimal()
```




```
train_no_anomaly <- train %>% filter(DAYS_EMPLOYED != 365243)

ggplot(train_no_anomaly, aes(x = DAYS_EMPLOYED)) +
  geom_histogram(bins = 50, fill = "#1F4E79") +
  labs(title = "Distribution of DAYS_EMPLOYED, excluding the 365243 value",
       x = "DAYS_EMPLOYED", y = "Count") +
  theme_minimal()
```



With the 365243 value included, the distribution shows a single large spike disconnected from the rest of the data; with that value excluded, the remaining values range from -17,912 to 0 (about 49 years to 0 years), consistent with a negative day-offset representing employment tenure.

Question 2: EXT_SOURCE Completeness and TARGET

```
train_ext <- train %>%
  mutate(ext_source_count = 3 - (is.na(EXT_SOURCE_1) + is.na(EXT_SOURCE_2) + is.na(E

train_ext %>%
  group_by(ext_source_count) %>%
  summarise(n = n(), pct_of_applicants = n() / nrow(train_ext) * 100, target_rate =
```

```
# A tibble: 4 x 4
  ext_source_count      n pct_of_applicants target_rate
             <dbl>  <int>             <dbl>       <dbl>
1                0    172            0.0559        8.14
2                1  36737           11.9           9.92
```

```
3                2 161013           52.4           8.18
4                3 109589           35.6           7.30
```

Applicants with all three EXT_SOURCE scores present (35.6% of applicants) have a TARGET rate of 7.30%; applicants with exactly one score present (11.9% of applicants) have the highest TARGET rate of the four groups at 9.92%; applicants missing all three EXT_SOURCE scores (0.06% of applicants, n=172) have a TARGET rate of 8.14%, close to the overall average of 8.07% and not the extreme seen in the one-score group.

Question 3: TARGET Rate for Applicants Missing All Three EXT_SOURCE Scores

```
train_ext2 <- train_ext %>%
  mutate(all_ext_missing = ext_source_count == 0)

tbl3 <- train_ext2 %>%
  group_by(all_ext_missing) %>%
  summarise(n = n(), target_rate = mean(TARGET) * 100)

tbl3
```

```
# A tibble: 2 x 3
  all_ext_missing      n target_rate
  <lgl>            <int>       <dbl>
1 FALSE           307339        8.07
2 TRUE               172        8.14
```

```
chisq.test(table(train_ext2$all_ext_missing, train_ext2$TARGET))
```

```

    Pearson's Chi-squared test with Yates' continuity correction

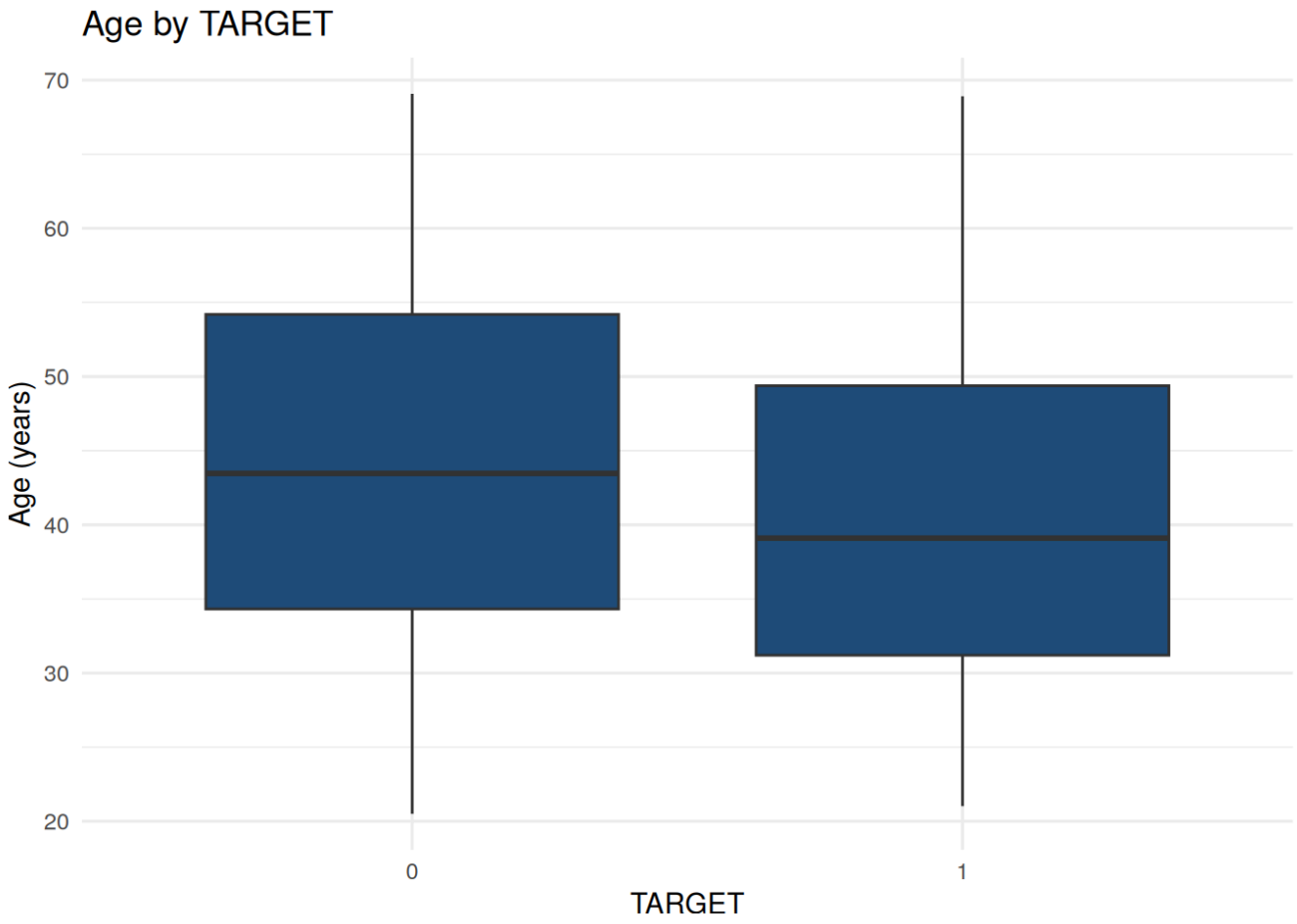
data:  table(train_ext2$all_ext_missing, train_ext2$TARGET)
X-squared = 2.0608e-27, df = 1, p-value = 1
```

Applicants missing all three EXT_SOURCE scores (172 applicants) have a TARGET rate of 8.14% compared to 8.07% for all other applicants (307,339 applicants), and a chi-square test comparing the two groups returns a p-value of 1, indicating no detectable difference between the groups.

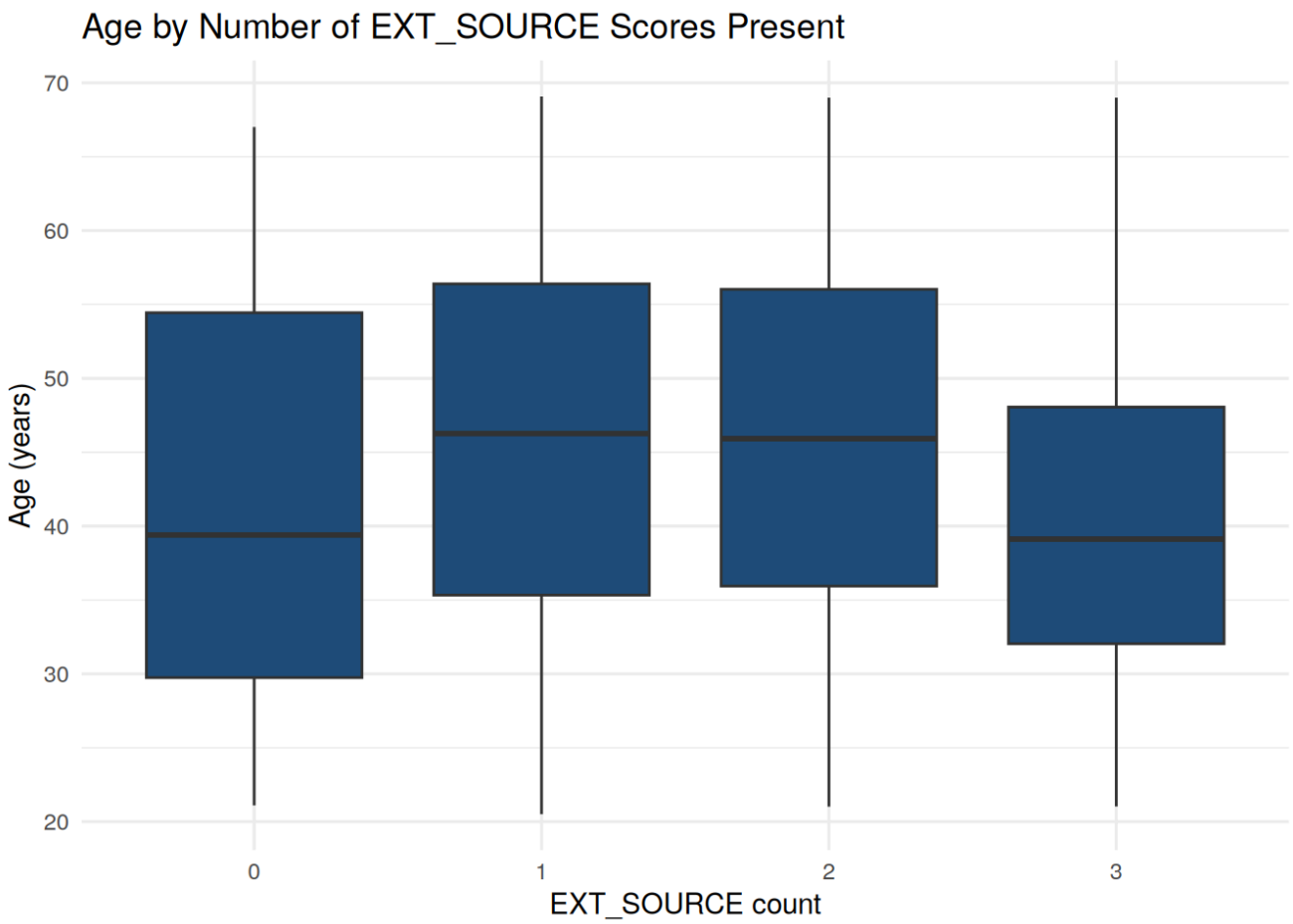
Question 9: Age vs. TARGET and vs. EXT_SOURCE Completeness

```r
train_age <- train_ext %>% mutate(age_years = -DAYS_BIRTH / 365.25)

ggplot(train_age, aes(x = factor(TARGET), y = age_years)) +
  geom_boxplot(fill = "#1F4E79") +
  labs(title = "Age by TARGET", x = "TARGET", y = "Age (years)") +
  theme_minimal()
```




```r
ggplot(train_age, aes(x = factor(ext_source_count), y = age_years)) +
  geom_boxplot(fill = "#1F4E79") +
  labs(title = "Age by Number of EXT_SOURCE Scores Present", x = "EXT_SOURCE count",
  theme_minimal()
```




```
train_age %>% group_by(TARGET) %>% summarise(median_age = median(age_years))
```

```
# A tibble: 2 x 2
  TARGET median_age
   <int>      <dbl>
1      0       43.5
2      1       39.1
```

```
train_age %>% group_by(ext_source_count) %>% summarise(median_age = median(age_years
```

```
# A tibble: 4 x 2
  ext_source_count median_age
             <dbl>      <dbl>
1                0       39.4
2                1       46.3
3                2       45.9
4                3       39.1
```

Median age is 39.1 years for TARGET=1 applicants and 43.5 years for TARGET=0 applicants; median age by EXT_SOURCE-present count is not monotonic — 39.4 years for 0 scores present, 46.3 years for 1 score present, 45.9 years for 2 scores present, and 39.1 years for 3

scores present, meaning the applicants with zero and three scores present are both younger than those with one or two present.

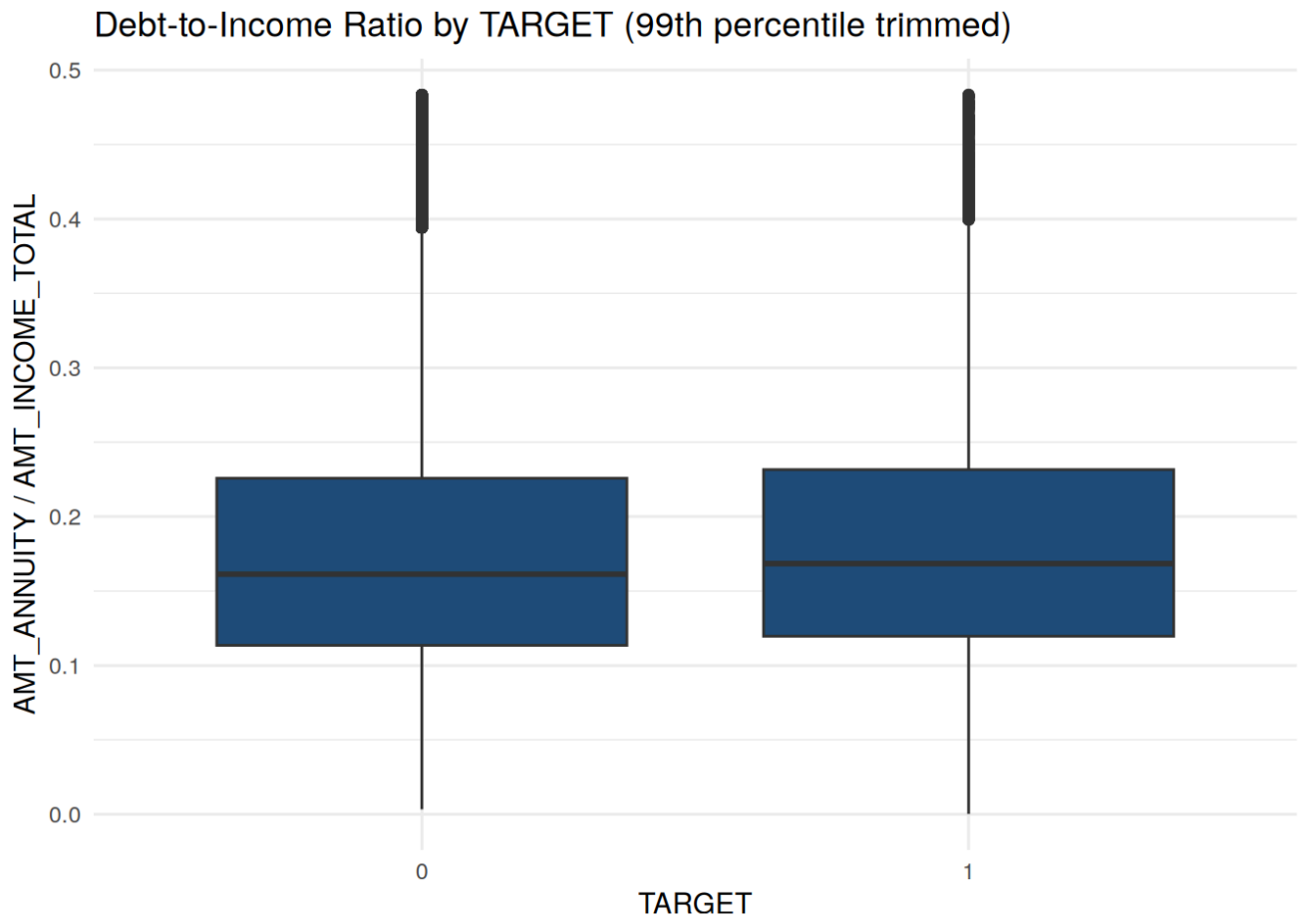
Question 4: Debt-to-Income Ratio vs. TARGET

```
train_ratios <- train %>%
  mutate(dti = AMT_ANNUITY / AMT_INCOME_TOTAL)

train_ratios %>% group_by(TARGET) %>% summarise(median_dti = median(dti, na.rm = TRU
```

```
# A tibble: 2 x 3
  TARGET median_dti mean_dti
   <int>      <dbl>    <dbl>
1      0      0.162    0.181
2      1      0.169    0.185
```

```
ggplot(train_ratios %>% filter(dti < quantile(dti, 0.99, na.rm = TRUE)), aes(x = fac
  geom_boxplot(fill = "#1F4E79") +
  labs(title = "Debt-to-Income Ratio by TARGET (99th percentile trimmed)", x = "TARG
  theme_minimal()
```



Median debt-to-income ratio is 0.169 for TARGET=1 applicants and 0.162 for TARGET=0 applicants; mean DTI is 0.185 vs. 0.181 respectively.

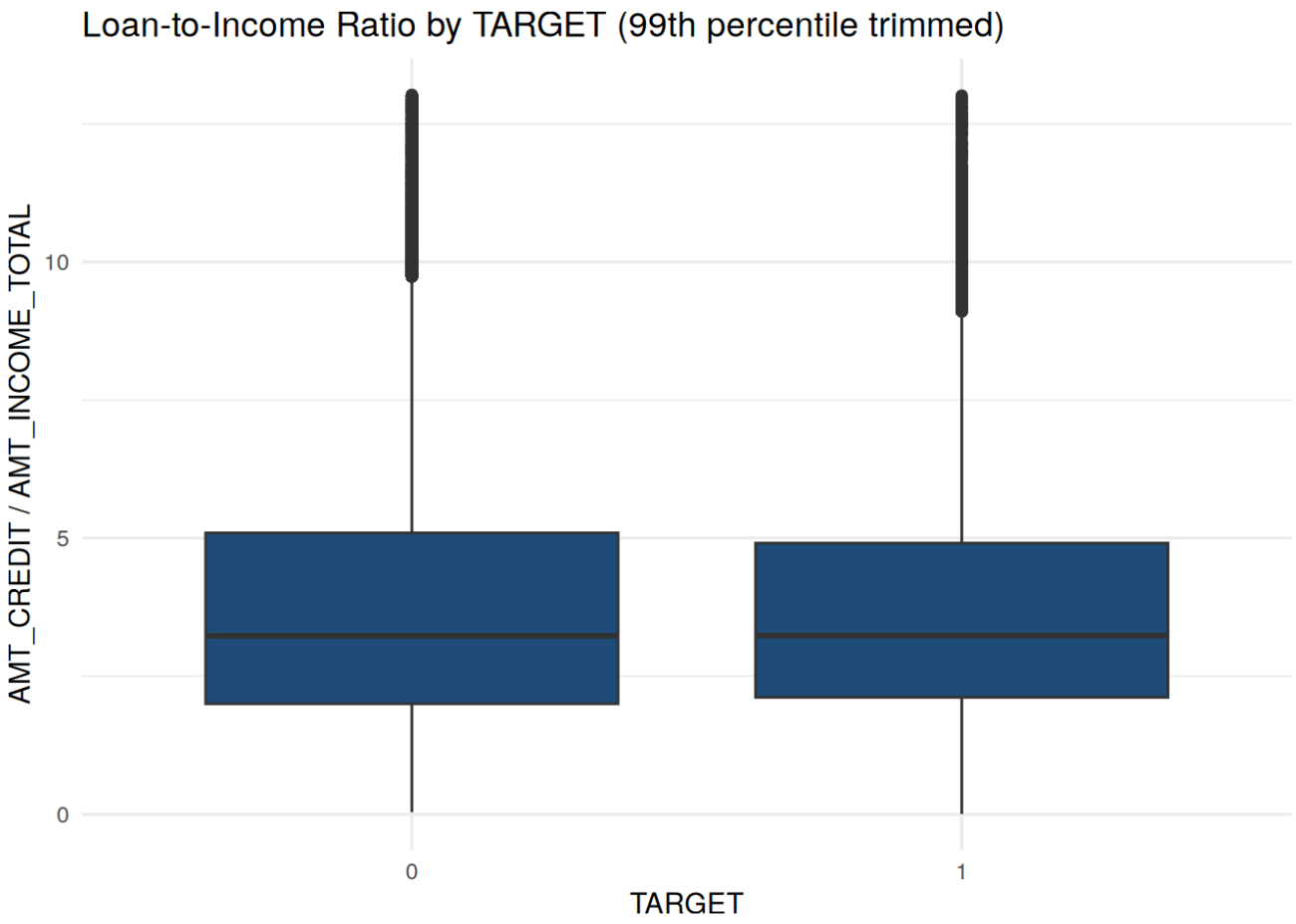
Question 5: Loan-to-Income Ratio vs. TARGET

```
train_ratios2 <- train_ratios %>%
  mutate(lti = AMT_CREDIT / AMT_INCOME_TOTAL)

train_ratios2 %>% group_by(TARGET) %>% summarise(median_lti = median(lti, na.rm = TR
```

```
# A tibble: 2 x 3
  TARGET median_lti mean_lti
   <int>      <dbl>    <dbl>
1      0       3.27     3.96
2      1       3.25     3.89
```

```
ggplot(train_ratios2 %>% filter(lti < quantile(lti, 0.99, na.rm = TRUE)), aes(x = fa
  geom_boxplot(fill = "#1F4E79") +
  labs(title = "Loan-to-Income Ratio by TARGET (99th percentile trimmed)", x = "TARG
  theme_minimal()
```



Median loan-to-income ratio is 3.25 for TARGET=1 applicants and 3.27 for TARGET=0 applicants — essentially the same, with TARGET=1 slightly lower rather than higher; mean LTI is 3.89 vs. 3.96 respectively.

Question 6: Loan-to-Value Ratio vs. TARGET, and AMT_GOODS_PRICE Missingness

```
mean(is.na(train$AMT_GOODS_PRICE)) * 100
```

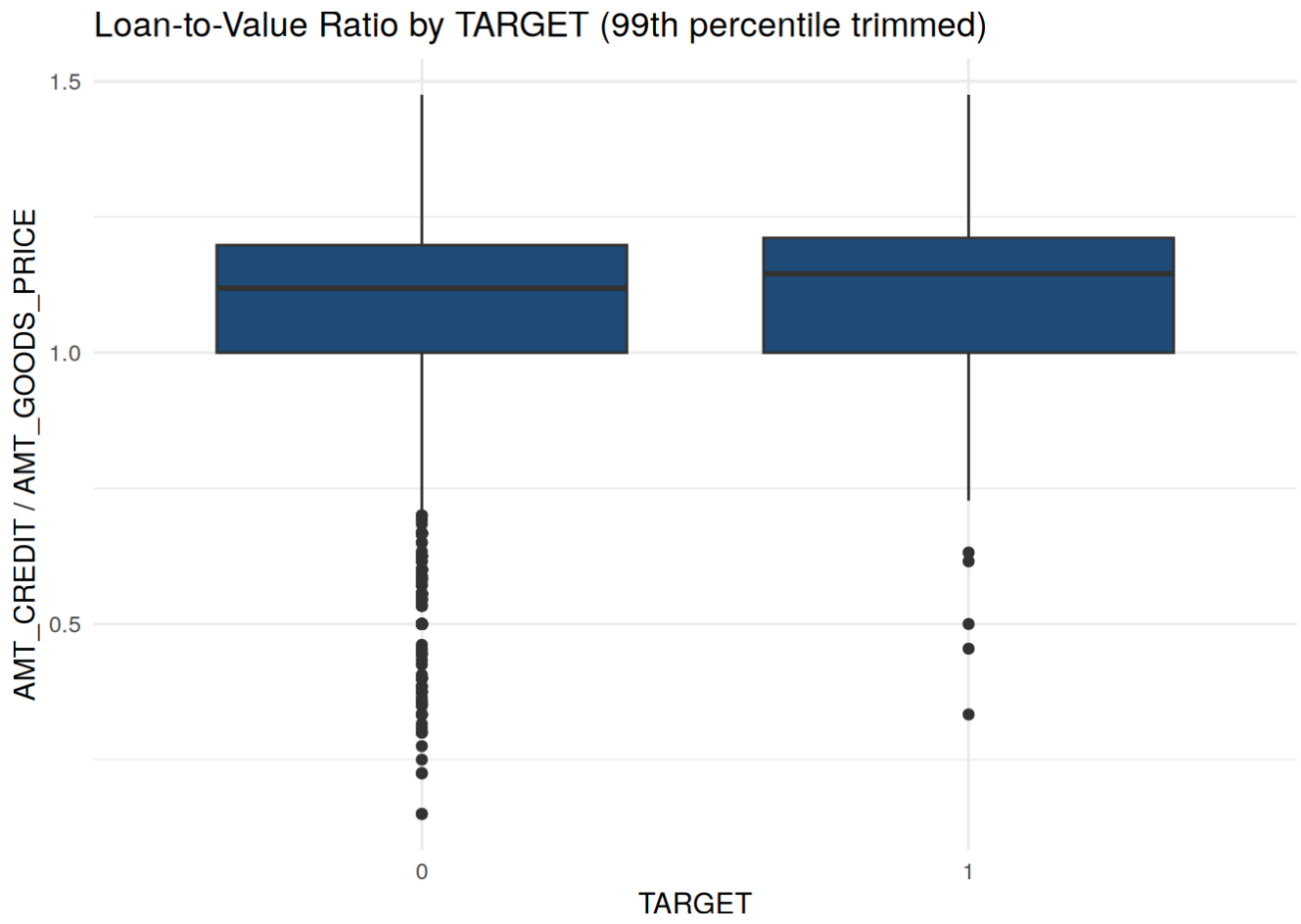
```
[1] 0.09040327
```

```
train_ltv <- train %>%
  filter(!is.na(AMT_GOODS_PRICE)) %>%
  mutate(ltv = AMT_CREDIT / AMT_GOODS_PRICE)

train_ltv %>% group_by(TARGET) %>% summarise(median_ltv = median(ltv), mean_ltv = me
```

```
# A tibble: 2 x 3
  TARGET median_ltv mean_ltv
   <int>      <dbl>    <dbl>
1      0       1.12     1.12
2      1       1.15     1.15
```

```
ggplot(train_ltv %>% filter(ltv < quantile(ltv, 0.99)), aes(x = factor(TARGET), y =
  geom_boxplot(fill = "#1F4E79") +
  labs(title = "Loan-to-Value Ratio by TARGET (99th percentile trimmed)", x = "TARGE
  theme_minimal()
```



`AMT_GOODS_PRICE` is missing for 0.09% of applicants; among applicants where it is present, median loan-to-value is 1.12 for TARGET=0 and 1.15 for TARGET=1 — both close to but not exactly 1.0.

Question 7: Credit Bureau Inquiries vs. TARGET

```
inquiry_cols <- c("AMT_REQ_CREDIT_BUREAU_HOUR", "AMT_REQ_CREDIT_BUREAU_DAY", "AMT_RE
                  "AMT_REQ_CREDIT_BUREAU_MON", "AMT_REQ_CREDIT_BUREAU_QRT", "AMT_RE

train %>%
  select(all_of(inquiry_cols), TARGET) %>%
  group_by(TARGET) %>%
  summarise(across(all_of(inquiry_cols), ~ mean(.x, na.rm = TRUE)))
```

```
# A tibble: 2 x 7
  TARGET AMT_REQ_CREDIT_BUREAU_H~1 AMT_REQ_CREDIT_BUREA~2 AMT_REQ_CREDIT_BUREA~3
   <int>                     <dbl>                  <dbl>                  <dbl>
1      0                   0.00638                0.00691                 0.0343
2      1                   0.00667                0.00804                 0.0349
# i abbreviated names: 1: AMT_REQ_CREDIT_BUREAU_HOUR,
#   2: AMT_REQ_CREDIT_BUREAU_DAY, 3: AMT_REQ_CREDIT_BUREAU_WEEK
```

```
# i 3 more variables: AMT_REQ_CREDIT_BUREAU_MON <dbl>,
#   AMT_REQ_CREDIT_BUREAU_QRT <dbl>, AMT_REQ_CREDIT_BUREAU_YEAR <dbl>
```

Results are mixed across the six inquiry-count windows: HOUR, DAY, WEEK, and YEAR are all higher for TARGET=1 than TARGET=0 (largest relative gap in DAY: 0.00804 vs. 0.00691, about 16% higher), while MON and QRT are actually lower for TARGET=1 (MON: 0.228 vs. 0.271, about 16% lower) — there is no consistent direction across all six windows.

Question 10: Distribution of Protected/Quasi-Protected Characteristics by TARGET (Descriptive)

```
train %>% group_by(CODE_GENDER) %>% summarise(n = n(), target_rate = mean(TARGET) *
```

```
# A tibble: 3 x 3
  CODE_GENDER      n target_rate
  <chr>        <int>       <dbl>
1 F           202448        7.00
2 M           105059       10.1
3 XNA              4        0
```

```
train %>% group_by(NAME_FAMILY_STATUS) %>% summarise(n = n(), target_rate = mean(TAR
```

```
# A tibble: 6 x 3
  NAME_FAMILY_STATUS        n target_rate
  <chr>                 <int>       <dbl>
1 Unknown                   2        0
2 Widow                 16088        5.82
3 Married              196432        7.56
4 Separated             19770        8.19
5 Single / not married  45444        9.81
6 Civil marriage        29775        9.94
```

```
train %>% group_by(CNT_CHILDREN) %>% summarise(n = n(), target_rate = mean(TARGET) *
```

```
# A tibble: 6 x 3
  CNT_CHILDREN      n target_rate
         <int>  <int>       <dbl>
1            0 215371        7.71
2            1  61119        8.92
3            2  26749        8.72
4            3   3717        9.63
5            4    429       12.8
6            5     84        8.33
```

TARGET rate by `CODE_GENDER` is 7.00% for F, 10.1% for M, and 0% for the 4 rows coded XNA; TARGET rate by `NAME_FAMILY_STATUS` ranges from 5.82% (Widow) to 9.94% (Civil marriage), with a 2-row "Unknown" category at 0%; TARGET rate by `CNT_CHILDREN` for the six most common values ranges from 7.71% (0 children, the largest group) up to 12.8% for the smaller group with 4 children.

Question 11: Train/Test Consistency for Key Numeric Columns

```
compare_cols <- c("AMT_INCOME_TOTAL", "AMT_CREDIT", "EXT_SOURCE_1", "EXT_SOURCE_2",

summary_stats <- function(df, label) {
  df %>%
    select(all_of(compare_cols)) %>%
    summarise(across(everything(), list(mean = ~mean(.x, na.rm = TRUE), median = ~me
    mutate(dataset = label)
}

rbind(summary_stats(train, "train"), summary_stats(test, "test"))
```

```
  AMT_INCOME_TOTAL_mean AMT_INCOME_TOTAL_median AMT_CREDIT_mean
1              168797.9                  147150        599026.0
2              178431.8                  157500        516740.4
  AMT_CREDIT_median EXT_SOURCE_1_mean EXT_SOURCE_1_median EXT_SOURCE_2_mean
1            513531         0.5021298           0.5059979         0.5143927
2            450000         0.5011798           0.5067713         0.5180211
  EXT_SOURCE_2_median EXT_SOURCE_3_mean EXT_SOURCE_3_median dataset
1           0.5659614         0.5108529           0.5352763   train
2           0.5587579         0.5001056           0.5190973    test
```

The three `EXT_SOURCE` columns are close between train and test (within about 2% on mean and median). `AMT_INCOME_TOTAL` differs by about 6% on mean and 7% on median (test higher). `AMT_CREDIT` differs substantially more — about 14% lower mean and 12% lower median in test than in train — which is a real distributional difference between the two populations, not a negligible one.

Is There Anything We Have Missed?

- **Against the BPS:** The BPS's claim that "false approvals and false declines have different costs" (Question 1 in the original question list) could not be tested — `application_train`/`application_test` contain only originated loans, so declined applicants and their would-have-been outcomes are not observable in this data at all. This is a material gap for the business problem statement's framing, not just a data note.

- **Against the dictionary:** No data dictionary was available for this pass (also flagged in `data_map.md` and `feature_report.md`). Several findings above — the `DAYS_EMPLOYED` sentinel value, the `CODE_GENDER` "XNA" category, and the exact definition of the `EXT_SOURCE` columns — would benefit from dictionary confirmation rather than inference from the data alone.
- **Against the standing checks:** All six were performed. One follow-up: S5 (temporal direction) could only be assessed by column-naming convention, not verified against the dictionary's stated timing for each field.
- **What an underwriter would refuse to believe:** A `DAYS_EMPLOYED` value of 365243 (about 1,000 years) is not physically possible and would need to be treated as a missing-value code, not a real tenure, before any modeling. The near-1.0 loan-to-value ratio for the median applicant (Question 6) is also worth an underwriter's skepticism — it suggests `AMT_GOODS_PRICE` may often be set close to `AMT_CREDIT` by convention rather than reflecting an independently priced good, which would make the loan-to-value ratio less informative than its name implies. Also worth flagging: several of the simple bivariate relationships expected to show a clear pattern (EXT_SOURCE-missing default rate in Question 3, loan-to-income in Question 5) instead showed no meaningful difference or even a reversed direction from what a simple "more risk factors = more default" story would predict — an underwriter reviewing this would reasonably ask whether these variables interact with others rather than acting independently, which single-variable EDA cannot resolve.

Results Summary

(This section is intentionally left for the person completing the analysis to write, per this project's workflow — interpretation belongs in `decisions.md`, not in this draft notebook.)

Appendix: Use of AI

(To be completed — see the note in the chat accompanying this notebook.)